SYED SABIRA
192373044

Below are **detailed, simple, human-written answers** for your **Scenario-Based Questions**. These are suitable for **5-mark university answers**.

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1. Use Case Diagram Scenario – Smart Pharmacy Management System

Scenario Analysis

A Smart Pharmacy Management System helps customers search for medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions before dispensing medicines, while administrators manage medicines and user accounts. The interactions between different users must be clearly identified to ensure smooth system operation.

Actors

- Customer
- Pharmacist
- Administrator
- Payment Gateway (External Actor)

Use Cases

- Register/Login
- Search Medicine
- Upload Prescription
- View Medicine Details
- Place Order
- Make Payment
- Verify Prescription
- Dispense Medicine
- Manage Inventory
- Manage Users
- Generate Reports

Relationships

Include Relationship

The **Place Order** use case includes **Upload Prescription** (for prescription medicines) and **Make Payment** because both are required to complete an order.

Extend Relationship

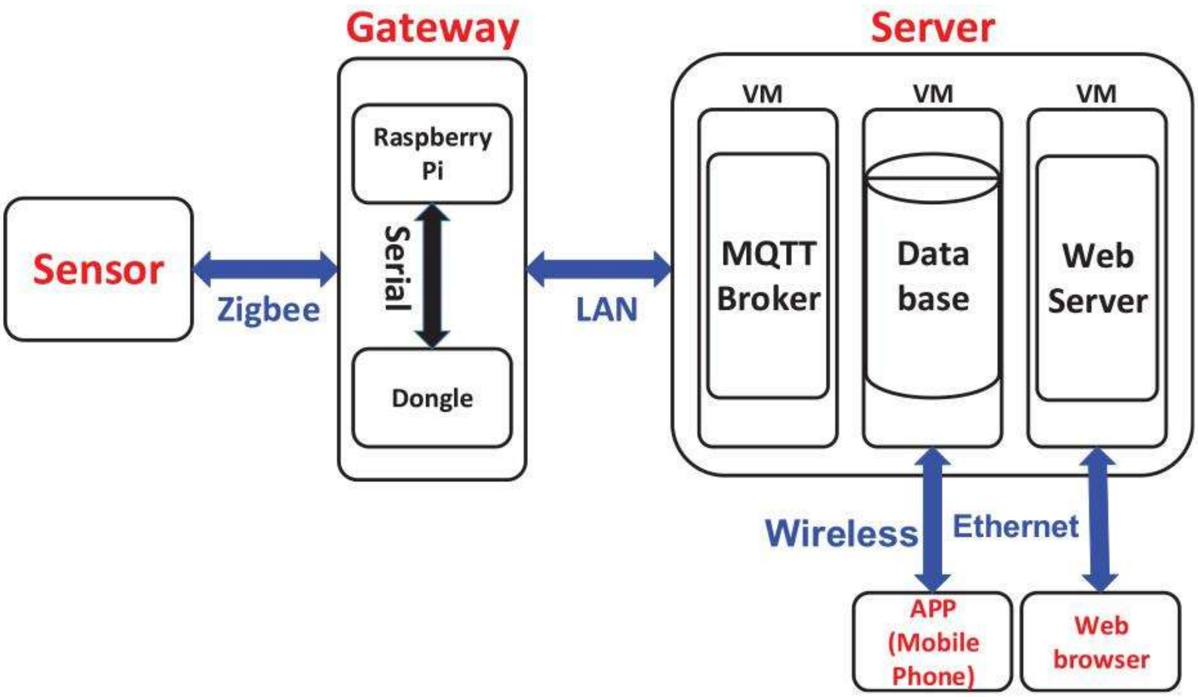
The **Verify Prescription** use case extends **Place Order** because verification is required only when prescription medicines are ordered.

Generalization

The **Administrator** and **Pharmacist** are specialized users of the system and inherit common login functionality.

Benefits of the System

- Enables online medicine ordering.
- Ensures only valid prescriptions are accepted.
- Improves medicine inventory management.
- Provides secure online payment.
- Reduces manual work and improves efficiency.



2. Class Diagram Scenario – Smart Hotel Management System

Scenario Analysis

A Smart Hotel Management System manages hotel operations such as guest registration, room allocation, reservations, payments, and receptionist activities. The system should clearly define classes, attributes, methods, and relationships among all entities.

Classes

Guest

Attributes

- Guest ID
- Name
- Phone Number
- Address

Methods

- Book Room()
- Cancel Reservation()
- Make Payment()

Room

Attributes

- Room Number
- Room Type
- Price
- Availability

Methods

- Check Availability()
- Update Status()

Reservation

Attributes

- Reservation ID
- Check-in Date
- Check-out Date
- Reservation Status

Methods

- Confirm Booking()
- Cancel Booking()

Payment

Attributes

- Payment ID
- Amount
- Payment Method
- Payment Status

Methods

- Process Payment()
- Generate Receipt()

Receptionist

Attributes

- Employee ID
- Name
- Contact Number

Methods

- Register Guest()
- Assign Room()
- Check Out Guest()

Relationships

Association

A Guest makes one or more Reservations.

Aggregation

A Hotel contains many Rooms.

Composition

A Reservation contains one Payment. If the reservation is deleted, the payment record is also removed.

Inheritance

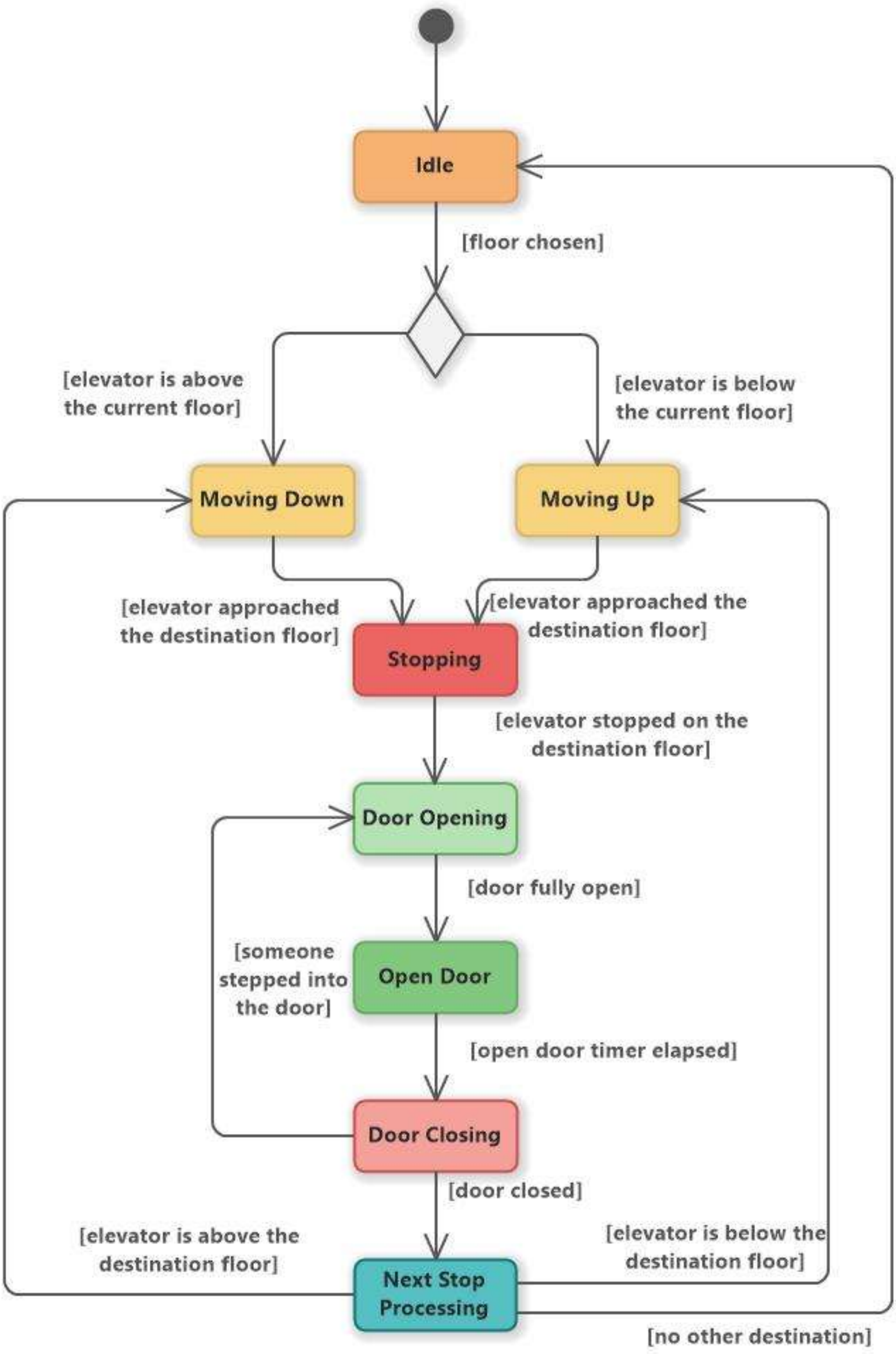
Manager and Receptionist can inherit common properties from the Employee class.

Multiplicity

- One Guest can make many Reservations.
- One Reservation is assigned to one Room.
- One Receptionist manages many Reservations.

Advantages

- Easy room booking.
- Better customer management.
- Efficient payment processing.
- Easy maintenance and future expansion.



3. Sequence Diagram Scenario – Online Banking System

Scenario Analysis

The Online Banking System allows customers to log in, transfer funds, verify transactions using OTP, and receive confirmation messages. Multiple system components communicate to complete the transaction securely.

Objects

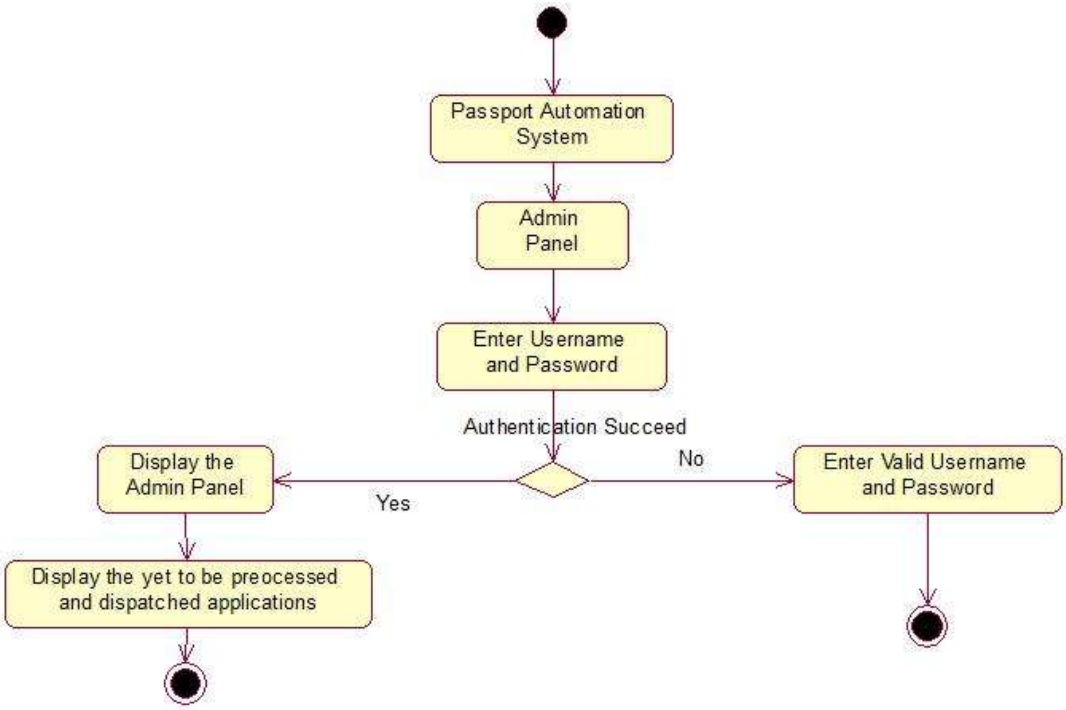
- Customer
- Login Page
- Authentication Server
- Banking Server
- Database
- OTP Service

Sequence of Operations

1. Customer enters username and password.
2. Login Page sends login request to Authentication Server.
3. Authentication Server validates credentials using the Database.
4. Authentication Server sends successful login response.
5. Customer selects Transfer Funds.
6. Banking Server requests OTP generation.
7. OTP Service sends OTP to customer.
8. Customer enters OTP.
9. Banking Server verifies OTP.
10. Banking Server updates the Database.
11. Database confirms successful transaction.
12. Banking Server sends confirmation message to the customer.

Advantages

- Secure transaction using OTP.
- Accurate fund transfer.
- Prevents unauthorized access.
- Fast transaction confirmation.
- Reliable banking service.



4. Activity Diagram Scenario – Passport Application System

Scenario Analysis

The Passport Application System allows applicants to submit passport applications, upload required documents, verify identity, pay processing fees, and receive passports. The workflow includes approval, rejection, and parallel processing activities.

Workflow

1. Applicant opens Passport Portal.
2. Fills the application form.
3. Uploads required documents.
4. System verifies document completeness.
5. Applicant pays processing fee.
6. Identity verification is performed.
7. Background verification is conducted simultaneously.
8. Both verification results are combined.
9. Decision is taken.

If Approved

- Passport is printed.
- Passport is dispatched.
- Applicant receives passport.

If Rejected

- Applicant receives rejection notification.
- Process ends.

Decision Points

- Documents valid?
- Fee payment successful?
- Identity verification successful?
- Final approval?

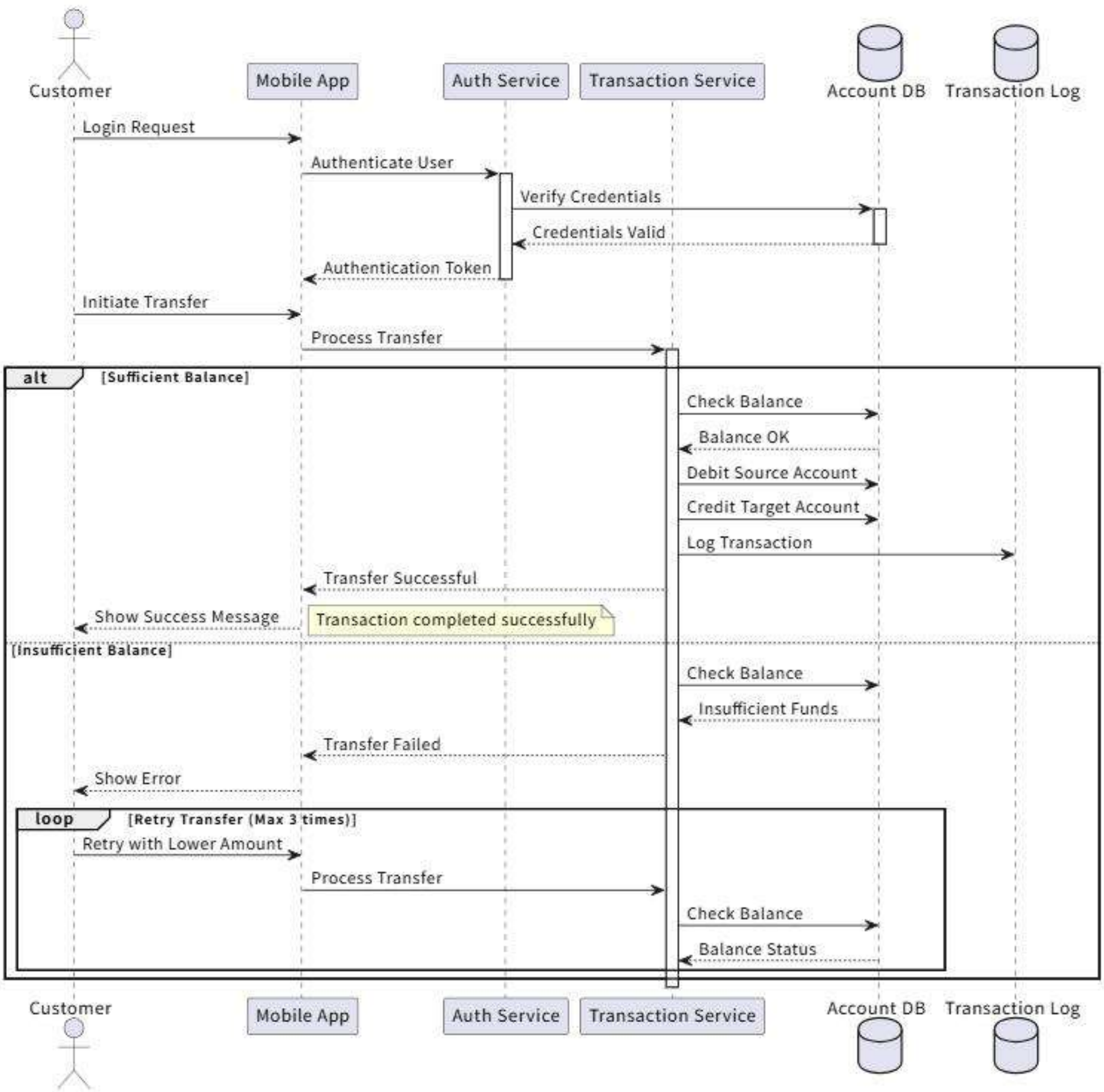
Parallel Activities

- Identity Verification
- Background Verification

Both activities are executed simultaneously and joined before the final approval.

Benefits

- Faster passport processing.
- Reduced paperwork.
- Better workflow management.
- Improved transparency.
- Efficient verification process.



5. State Diagram Scenario – Smart Elevator Control System

Scenario Analysis

A Smart Elevator Control System changes its behavior depending on user requests, floor selection, emergencies, and maintenance activities. The elevator moves between different operational states.

States

- Idle
- Door Open
- Door Closed
- Moving Up
- Moving Down
- Emergency Stop

- Maintenance Mode

Transitions

- Elevator starts in **Idle**.
- User presses elevator button.
- Elevator changes to **Door Open**.
- Passenger enters.
- Door closes.
- Elevator enters **Door Closed** state.
- If destination floor is above, elevator moves to **Moving Up**.
- If destination floor is below, elevator moves to **Moving Down**.
- Elevator reaches destination.
- Door opens.
- Passengers exit.
- Elevator returns to **Idle**.

Emergency Event

If emergency button is pressed, elevator immediately changes to **Emergency Stop**.

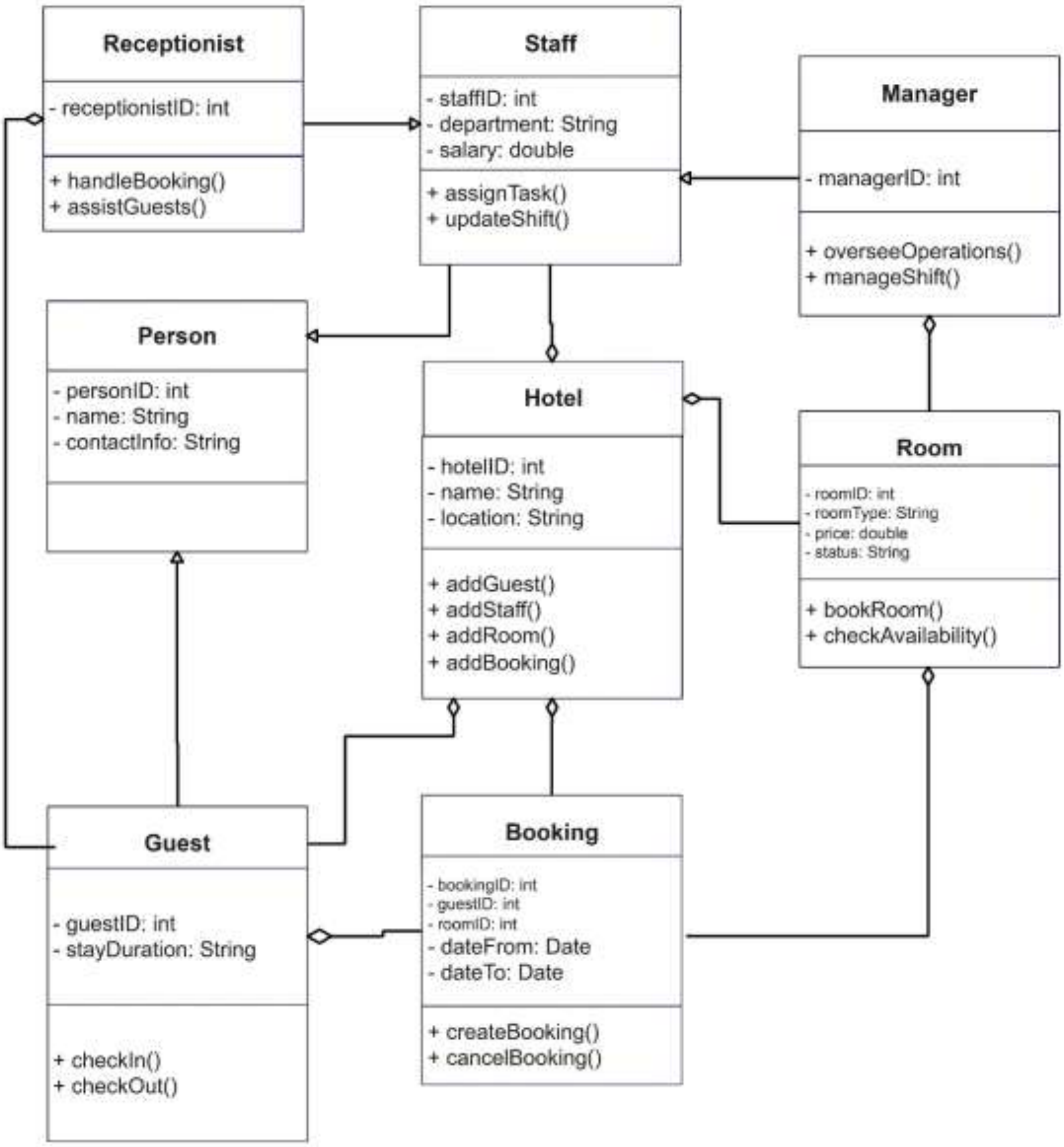
Maintenance Event

If maintenance mode is activated, elevator enters **Maintenance Mode** and does not respond to user requests.

Advantages

- Safe elevator operation.
- Efficient movement between floors.
- Handles emergency situations.
- Supports maintenance activities.
- Improves passenger safety.

Class Diagram for Hotel Management



6. Deployment Diagram Scenario – Smart Home Automation System

Scenario Analysis

A Smart Home Automation System connects smart sensors, IoT controllers, a home gateway, cloud servers, and a mobile application. The system allows users to monitor and control lighting, security, and appliances remotely.

Hardware Nodes

- Mobile Phone
- Smart Sensors

- IoT Controller
- Home Gateway
- Wi-Fi Router
- Cloud Server

Software Components

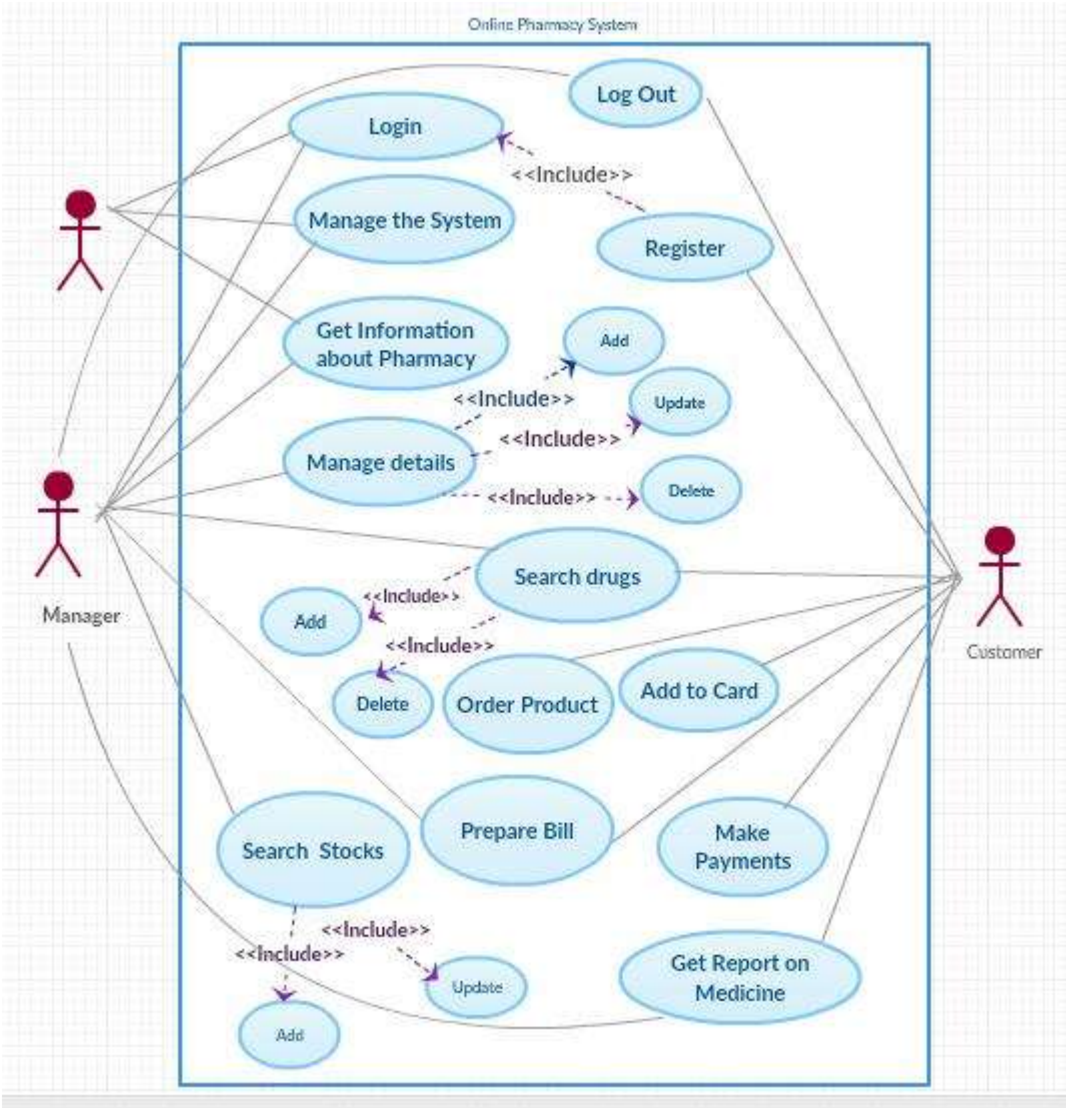
- Mobile Application
- Home Automation Software
- Sensor Monitoring Service
- Cloud Database
- Notification Service

Communication

1. Smart Sensors collect environmental data.
2. Sensors send data to the IoT Controller.
3. IoT Controller forwards data to the Home Gateway.
4. Home Gateway communicates with the Cloud Server through the Internet.
5. Cloud Server stores data in the Cloud Database.
6. User accesses the Mobile Application.
7. Mobile App sends control commands to the Cloud Server.
8. Cloud Server forwards commands to the Home Gateway.
9. Home Gateway sends commands to the IoT Controller.
10. IoT Controller controls lights, security systems, and appliances.

Advantages

- Remote monitoring and control.
- Real-time notifications.
- Secure cloud storage.
- Easy device management.
- Energy-efficient home automation.
- Scalable and reliable architecture.


Online Pharmacy System
Log Out
Login
<<Include>>
Manage the System
Register
Get Information about Pharmacy
Add
Update
Delete
Manage details
<<Include>>
Manager
Customer
Search drugs
Add
Delete
Order Product
Add to Card
Search Stocks
Prepare Bill
Make Payments
Update
Add
Get Report on Medicine